This is what run-length encoding requires. Correspondingly, delta encoding followed by Huffman and/or run-length encoding is a common strategy for compressing signals.

The idea used in delta encoding can be expanded into a more complicated technique called **Linear Predictive Coding**, or **LPC**. To understand LPC, imagine that the first 99 samples from the input signal have been encoded, and we are about to work on sample number 100. We then ask ourselves: based on the first 99 samples, what is the most likely value for sample 100? In delta encoding, the answer is that the most likely value for sample 100 is the same as the previous value, sample 99. This expected value is used as a reference to encode sample 100. That is, the *difference* between the sample and the expectation is placed in the encoded file. LPC expands on this by making a better guess at what the most probable value is. This is done by looking at the last several samples, rather than just the last sample. The algorithms used by LPC are similar to recursive filters, making use of the z-transform and other intensively mathematical techniques.

LZW Compression

LZW compression is named after its developers, A. Lempel and J. Ziv, with later modifications by Terry A. Welch. It is the foremost technique for general purpose data compression due to its simplicity and versatility. Typically, you can expect LZW to compress text, executable code, and similar data files to about one-half their original size. LZW also performs well when presented with extremely redundant data files, such as tabulated numbers, computer source code, and acquired signals. Compression ratios of 5:1 are common for these cases. LZW is the basis of several personal computer utilities that claim to *"double the capacity of your hard drive."*

LZW compression is always used in GIF image files, and offered as an option in TIFF and PostScript. LZW compression is protected under U.S. patent number 4,558,302, granted December 10, 1985 to Sperry Corporation (now the Unisys Corporation). For information on commercial licensing, contact: Welch Licensing Department, Law Department, M/SC2SW1, Unisys Corporation, Blue Bell, Pennsylvania, 19424-0001.

LZW compression uses a **code table**, as illustrated in Fig. 27-6. A common choice is to provide 4096 entries in the table. In this case, the LZW encoded data consists entirely of 12 bit codes, each referring to one of the entries in the code table. Uncompression is achieved by taking each code from the compressed file, and translating it through the code table to find what character or characters it represents. Codes 0-255 in the code table are always assigned to represent single bytes from the input file. For example, if only these first 256 codes were used, each byte in the original file would be converted into 12 bits in the LZW encoded file, resulting in a 50% larger file size. During uncompression, each 12 bit code would be translated via the code table back into the single bytes. Of course, this wouldn't be a useful situation.